



7993 - Do protoplanetary disk substructures regulate water enrichment and pebble drift across different environments?

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	SO1036 observations	MIRI Medium Resolution Spectroscopy	(1) SO1036
	2	SO1153 observations	MIRI Medium Resolution Spectroscopy	(2) SO1153
	3	SO897 observations	MIRI Medium Resolution Spectroscopy	(3) SO897
	4	SO844 observations	MIRI Medium Resolution Spectroscopy	(4) SO844
	5	SO1152 observations	MIRI Medium Resolution Spectroscopy	(5) SO1152
	6	SO1274 observations	MIRI Medium Resolution Spectroscopy	(6) SO1274
	7	SO662 observations	MIRI Medium Resolution Spectroscopy	(7) SO662
	8	SO984 observations	MIRI Medium Resolution Spectroscopy	(8) SO984
	10	Athalia Observations	MIRI Medium Resolution Spectroscopy	(10) 515-Athalia

ABSTRACT

The inward radial transport of icy pebbles in protoplanetary disks influences the masses, compositions, and habitability of forming planets. Observations of disks in nearby star-forming regions have shown that the cool water vapor content in the inner disk and inferred pebble flux are anticorrelated with disk size, a trend that has been ascribed to larger disks having prominent substructures that trap dust and reduce the icy pebble flux to the inner disk. Planet-disk interactions are the leading explanation for disk substructures, suggesting that the formation of wide-separation giant planets regulates whether the inner disk is water-rich or poor and thus the properties of planets that form there. However, the relationship between the inner disk water reservoir and outer disk structure has thus far only been studied in star-forming regions with mild UV fields. Most stars emerge in high UV environments, which can drive mass loss in the outer regions of disks through external photoevaporation and thus reduce pebble drift to the inner disk. To investigate whether the observed link between the inner disk water content and outer disk structure generalizes to more typical star-forming environments, we propose to obtain MIRI MRS spectra of disks in the irradiated Sigma Orionis cluster that have been imaged at high resolution with ALMA. These observations will shed light on the universality of processes that control the composition and architecture of planetary systems.

OBSERVING DESCRIPTION

This program will obtain MIRI MRS spectra of eight protoplanetary disks in the Sigma Ori cluster that have previously been imaged with ALMA at high resolution. All three grating settings will be used to provide wavelength coverage from 5 to 28 microns in order to target water emission lines across a range of upper state energy levels, along with other disk chemistry tracers such as CO₂, OH, HCN, and C₂H₂. The sources are expected to be unresolved, so the 4-point dither for point sources will be used for all observations. Exposure times are chosen to achieve an SNR of at least 100 at

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25 microns in order to have adequate sensitivity to water lines at lower energy levels, which probe the cooler water thought to be associated with pebble drift. Target acquisition will be used for all sources in order to facilitate fringe removal. Using the same setup, we will also obtain a spectrum of asteroid 515 Athalia with MIRI MRS in order to measure the relative spectral response function, which is necessary to optimize fringe removal in the spectra of the disk targets. We aim for an SNR of 300 at 25 microns, i.e., 3x higher than that of the science targets. By default, simultaneous imaging is included for all observations.

Proposal 7993 - Targets - Do protoplanetary disk substructures regulate water enrichment and pebble drift across different environme...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SO1036	RA: 05 39 25.2031 (84.8550129d) Dec: -02 38 22.06 (-2.63946d) Equinox: J2000	Proper Motion RA: 1.843 mas/yr Proper Motion Dec: -0.3490000608508126 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO				
	(2)	SO1153	RA: 05 39 39.8262 (84.9159425d) Dec: -02 31 21.87 (-2.52274d) Equinox: J2000	Proper Motion RA: 1.797999999999998 mas/yr Proper Motion Dec: -0.09400009730597958 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO				
	(3)	SO897	RA: 05 39 7.6049 (84.7816871d) Dec: -02 32 39.19 (-2.54422d) Equinox: J2000	Proper Motion RA: 1.588 mas/yr Proper Motion Dec: -0.9219999355991604 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO				
Fixed Targets	(4)	SO844	RA: 05 39 1.3710 (84.7557125d) Dec: -02 18 27.46 (-2.30763d) Equinox: J2000	Proper Motion RA: 1.704 mas/yr Proper Motion Dec: -0.05199992756388383 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO				
	(5)	SO1152	RA: 05 39 39.3781 (84.9140754d) Dec: -02 17 4.50 (-2.28458d) Equinox: J2000	Proper Motion RA: 2.105 mas/yr Proper Motion Dec: -0.03300006028439384 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO				
Fixed Targets	(6)	SO1274	RA: 05 39 54.6558 (84.9777325d) Dec: -02 46 34.13 (-2.77615d) Equinox: J2000	Proper Motion RA: 2.166 mas/yr Proper Motion Dec: -0.7809999942764989 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO				

Proposal 7993 - Targets - Do protoplanetary disk substructures regulate water enrichment and pebble drift across different environme...

Solar System Targets	(7)	SO662	RA: 05 38 40.2735 (84.6678062d) Dec: -02 30 18.56 (-2.50516d) Equinox: J2000	Proper Motion RA: 0.8710000000000001 mas/yr Proper Motion Dec: -0.6099999609432416 mas/yr Parallax: 0.0024925000000000004" Epoch of Position: 2000
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks, T Tauri stars, Young stellar objects] Extended=NO			
	(8)	SO984	RA: 05 39 18.8360 (84.8284833d) Dec: -02 30 53.22 (-2.51478d) Equinox: J2000	Proper Motion RA: 1.808 mas/yr Proper Motion Dec: -0.6540000413224334 mas/yr Parallax: 0.0024411999999999997" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks, T Tauri stars, Young stellar objects] Extended=NO				
Solar System Targets	#	Name	Level 1	Level 2
	(10)	515-Athalia	TYPE=ASTEROID,A=3.121438411947196,E=0.1748 033154999111,I=2.038077430512606 ,O=121.0979177766495,W=300.2240591131205,M=2 12.2617984081051,EQUINOX=J2000,EPOCH=11- MAR-2018:00:00:00,EpochTimeScale=TDB	Level 3
Comments: Extended=NO				

Proposal 7993 - Observation 1 - Do protoplanetary disk substructures regulate water enrichment and pebble drift across different enviro...

Observation	Proposal 7993, Observation 1: SO1036 observations												Mon Dec 01 19:01:30 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	SO1036	RA: 05 39 25.2031 (84.8550129d)				Proper Motion RA: 1.843 mas/yr						
			Dec: -02 38 22.06 (-2.63946d)				Proper Motion Dec: -0.3490000608508126 mas/yr						
			Equinox: J2000				Epoch of Position: 2000						
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO											
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	FAST	8	1		1	22.2	221962			
Template	Primary Channel		Simultaneous Imaging				Imager Subarray			Grating Wheel Direction			
	All MRS		YES				FULL			Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	5	10	1	Dither 1	4	40	654.909	221962
	1	SHORT(A)	MRSLONG		FASTR1	80	2	1	Dither 1	4	8	1787.126	221962
	1	SHORT(A)	MRSSHORT		FASTR1	80	2	1	Dither 1	4	8	1787.126	221962
	2		IMAGER	F770W	FASTR1	5	10	1	Dither 1	4	40	654.909	221962
	2	MEDIUM(B)	MRSLONG		FASTR1	80	2	1	Dither 1	4	8	1787.126	221962
	2	MEDIUM(B)	MRSSHORT		FASTR1	80	2	1	Dither 1	4	8	1787.126	221962
	3		IMAGER	F1000W	FASTR1	5	10	1	Dither 1	4	40	654.909	221962
	3	LONG(C)	MRSLONG		FASTR1	80	2	1	Dither 1	4	8	1787.126	221962
	3	LONG(C)	MRSSHORT		FASTR1	80	2	1	Dither 1	4	8	1787.126	221962

Proposal 7993 - Observation 2 - Do protoplanetary disk substructures regulate water enrichment and pebble drift across different enviro...

Observation	Proposal 7993, Observation 2: SO1153 observations												Mon Dec 01 19:01:30 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	SO1153	RA: 05 39 39.8262 (84.9159425d)				Proper Motion RA: 1.797999999999998 mas/yr						
			Dec: -02 31 21.87 (-2.52274d)				Proper Motion Dec: -0.09400009730597958 mas/yr						
			Equinox: J2000				Epoch of Position: 2000						
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO										
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	FAST	8	1		1	22.2	221962			
Template	Primary Channel		Simultaneous Imaging				Imager Subarray			Grating Wheel Direction			
	All MRS		YES				FULL			Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	5	7	1	Dither 1	4	28	455.107	221962
	1	SHORT(A)	MRSLONG		FASTR1	20	2	1	Dither 1	4	8	455.107	221962
	1	SHORT(A)	MRSSHORT		FASTR1	20	2	1	Dither 1	4	8	455.107	221962
	2		IMAGER	F770W	FASTR1	5	7	1	Dither 1	4	28	455.107	221962
	2	MEDIUM(B)	MRSLONG		FASTR1	20	2	1	Dither 1	4	8	455.107	221962
	2	MEDIUM(B)	MRSSHORT		FASTR1	20	2	1	Dither 1	4	8	455.107	221962
	3		IMAGER	F1000W	FASTR1	5	7	1	Dither 1	4	28	455.107	221962
	3	LONG(C)	MRSLONG		FASTR1	20	2	1	Dither 1	4	8	455.107	221962
	3	LONG(C)	MRSSHORT		FASTR1	20	2	1	Dither 1	4	8	455.107	221962

Proposal 7993 - Observation 3 - Do protoplanetary disk substructures regulate water enrichment and pebble drift across different enviro...

Observation	Proposal 7993, Observation 3: SO897 observations												Mon Dec 01 19:01:30 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(3)	SO897	RA: 05 39 7.6049 (84.7816871d)				Proper Motion RA: 1.588 mas/yr						
			Dec: -02 32 39.19 (-2.54422d)				Proper Motion Dec: -0.9219999355991604 mas/yr						
			Equinox: J2000				Epoch of Position: 2000						
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO										
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	FAST	8	1		1	22.2	221962			
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	5	5	1	Dither 1	4	20	321.905	221962
	1	SHORT(A)	MRSLONG		FASTR1	105	2	1	Dither 1	4	8	2342.134	221962
	1	SHORT(A)	MRSSHORT		FASTR1	105	2	1	Dither 1	4	8	2342.134	221962
	2		IMAGER	F770W	FASTR1	5	5	1	Dither 1	4	20	321.905	221962
	2	MEDIUM(B)	MRSLONG		FASTR1	105	2	1	Dither 1	4	8	2342.134	221962
	2	MEDIUM(B)	MRSSHORT		FASTR1	105	2	1	Dither 1	4	8	2342.134	221962
	3		IMAGER	F1000W	FASTR1	5	5	1	Dither 1	4	20	321.905	221962
	3	LONG(C)	MRSLONG		FASTR1	105	2	1	Dither 1	4	8	2342.134	221962
	3	LONG(C)	MRSSHORT		FASTR1	105	2	1	Dither 1	4	8	2342.134	221962

Proposal 7993 - Observation 4 - Do protoplanetary disk substructures regulate water enrichment and pebble drift across different enviro...

Observation	Proposal 7993, Observation 4: SO844 observations												Mon Dec 01 19:01:30 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(4)	SO844	RA: 05 39 1.3710 (84.7557125d)				Proper Motion RA: 1.704 mas/yr						
			Dec: -02 18 27.46 (-2.30763d)				Proper Motion Dec: -0.05199992756388383 mas/yr						
			Equinox: J2000				Epoch of Position: 2000						
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO											
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	FAST	8	1		1	22.2	221962			
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		YES				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	5	12	1	Dither 1	4	48	788.111	221962
	1	SHORT(A)	MRSLONG		FASTR1	45	2	1	Dither 1	4	8	1010.115	221962
	1	SHORT(A)	MRSSHORT		FASTR1	45	2	1	Dither 1	4	8	1010.115	221962
	2		IMAGER	F770W	FASTR1	5	12	1	Dither 1	4	48	788.111	221962
	2	MEDIUM(B)	MRSLONG		FASTR1	45	2	1	Dither 1	4	8	1010.115	221962
	2	MEDIUM(B)	MRSSHORT		FASTR1	45	2	1	Dither 1	4	8	1010.115	221962
	3		IMAGER	F1000W	FASTR1	5	12	1	Dither 1	4	48	788.111	221962
	3	LONG(C)	MRSLONG		FASTR1	45	2	1	Dither 1	4	8	1010.115	221962
	3	LONG(C)	MRSSHORT		FASTR1	45	2	1	Dither 1	4	8	1010.115	221962

Proposal 7993 - Observation 5 - Do protoplanetary disk substructures regulate water enrichment and pebble drift across different enviro...

Observation	Proposal 7993, Observation 5: SO1152 observations												Mon Dec 01 19:01:30 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(5)	SO1152	RA: 05 39 39.3781 (84.9140754d)				Proper Motion RA: 2.105 mas/yr						
			Dec: -02 17 4.50 (-2.28458d)				Proper Motion Dec: -0.03300006028439384 mas/yr						
			Equinox: J2000				Epoch of Position: 2000						
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO											
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	FND	FAST	8	1	1	22.2	221962				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray			Grating Wheel Direction			
	All MRS		YES				FULL			Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	5	8	1	Dither 1	4	32	521.708	221962
	1	SHORT(A)	MRSLONG		FASTR1	85	2	1	Dither 1	4	8	1898.127	221962
	1	SHORT(A)	MRSSHORT		FASTR1	85	2	1	Dither 1	4	8	1898.127	221962
	2		IMAGER	F770W	FASTR1	5	8	1	Dither 1	4	32	521.708	221962
	2	MEDIUM(B)	MRSLONG		FASTR1	85	2	1	Dither 1	4	8	1898.127	221962
	2	MEDIUM(B)	MRSSHORT		FASTR1	85	2	1	Dither 1	4	8	1898.127	221962
	3		IMAGER	F1000W	FASTR1	5	8	1	Dither 1	4	32	521.708	221962
	3	LONG(C)	MRSLONG		FASTR1	85	2	1	Dither 1	4	8	1898.127	221962
	3	LONG(C)	MRSSHORT		FASTR1	85	2	1	Dither 1	4	8	1898.127	221962

Proposal 7993 - Observation 6 - Do protoplanetary disk substructures regulate water enrichment and pebble drift across different enviro...

Observation	Proposal 7993, Observation 6: SO1274 observations												Mon Dec 01 19:01:30 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(6)	SO1274	RA: 05 39 54.6558 (84.9777325d)				Proper Motion RA: 2.166 mas/yr						
			Dec: -02 46 34.13 (-2.77615d)				Proper Motion Dec: -0.7809999942764989 mas/yr						
			Equinox: J2000				Epoch of Position: 2000						
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks] Extended=NO											
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	FND	FAST	8	1	1	22.2	221962				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	5	15	1	Dither 1	4	60	987.914	221962
	1	SHORT(A)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116	221962
	1	SHORT(A)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116	221962
	2		IMAGER	F770W	FASTR1	5	15	1	Dither 1	4	60	987.914	221962
	2	MEDIUM(B)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116	221962
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116	221962
	3		IMAGER	F1000W	FASTR1	5	15	1	Dither 1	4	60	987.914	221962
	3	LONG(C)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116	221962
	3	LONG(C)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116	221962

Proposal 7993 - Observation 7 - Do protoplanetary disk substructures regulate water enrichment and pebble drift across different enviro...

Observation	Proposal 7993, Observation 7: SO662 observations											Mon Dec 01 19:01:30 GMT 2025	
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous				
	(7)	SO662	RA: 05 38 40.2735 (84.6678062d) Dec: -02 30 18.56 (-2.50516d) Equinox: J2000			Proper Motion RA: 0.8710000000000001 mas/yr Proper Motion Dec: -0.6099999609432416 mas/yr Parallax: 0.0024925000000000004" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks, T Tauri stars, Young stellar objects] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	FND	FAST	8	1	1	22.2	221962				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	5	3	1	Dither 1	4	12	188.703	221962
	1	SHORT(A)	MRSLONG		FASTR1	95	2	1	Dither 1	4	8	2120.131	221962
	1	SHORT(A)	MRSSHORT		FASTR1	95	2	1	Dither 1	4	8	2120.131	221962
	2		IMAGER	F770W	FASTR1	5	3	1	Dither 1	4	12	188.703	221962
	2	MEDIUM(B)	MRSLONG		FASTR1	95	2	1	Dither 1	4	8	2120.131	221962
	2	MEDIUM(B)	MRSSHORT		FASTR1	95	2	1	Dither 1	4	8	2120.131	221962
	3		IMAGER	F1000W	FASTR1	5	3	1	Dither 1	4	12	188.703	221962
	3	LONG(C)	MRSLONG		FASTR1	95	2	1	Dither 1	4	8	2120.131	221962
	3	LONG(C)	MRSSHORT		FASTR1	95	2	1	Dither 1	4	8	2120.131	221962

Proposal 7993 - Observation 8 - Do protoplanetary disk substructures regulate water enrichment and pebble drift across different enviro...

Observation	Proposal 7993, Observation 8: SO984 observations											Mon Dec 01 19:01:30 GMT 2025	
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 8:1) Warning (Form): Data Excess over lower threshold												
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(8)	SO984	RA: 05 39 18.8360 (84.8284833d) Dec: -02 30 53.22 (-2.51478d) Equinox: J2000				Proper Motion RA: 1.808 mas/yr Proper Motion Dec: -0.6540000413224334 mas/yr Parallax: 0.002441199999999997" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks, T Tauri stars, Young stellar objects] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	FAST	8	1		1	22.2	221962			
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	5	10	1	Dither 1	4	40	654.909	221962
	1	SHORT(A)	MRSLONG		FASTR1	140	2	1	Dither 1	4	8	3119.145	221962
	1	SHORT(A)	MRSSHORT		FASTR1	140	2	1	Dither 1	4	8	3119.145	221962
	2		IMAGER	F770W	FASTR1	5	10	1	Dither 1	4	40	654.909	221962
	2	MEDIUM(B)	MRSLONG		FASTR1	140	2	1	Dither 1	4	8	3119.145	221962
	2	MEDIUM(B)	MRSSHORT		FASTR1	140	2	1	Dither 1	4	8	3119.145	221962
	3		IMAGER	F1000W	FASTR1	5	10	1	Dither 1	4	40	654.909	221962
	3	LONG(C)	MRSLONG		FASTR1	140	2	1	Dither 1	4	8	3119.145	221962
	3	LONG(C)	MRSSHORT		FASTR1	140	2	1	Dither 1	4	8	3119.145	221962

Proposal 7993 - Observation 10 - Do protoplanetary disk substructures regulate water enrichment and pebble drift across different env...

Observation	Proposal 7993, Observation 10: Athalia Observations												Mon Dec 01 19:01:30 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(Athalia Observations (Obs 10)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(10)	515-Athalia	TYPE=ASTEROID,A=3.121438411947196,E=0.1748 033154999111,I=2.038077430512606 ,O=121.0979177766495,W=300.2240591131205,M=2 12.2617984081051,EQUINOX=J2000,EPOCH=11- MAR-2018:00:00:00,EpochTimeScale=TDB										
	Comments: Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	FND	FAST	4	1	1	11.1	222197				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	5	10	1	Dither 1	4	40	654.909	222197
	1	SHORT(A)	MRSLONG		FASTR1	15	5	1	Dither 1	4	20	876.913	222197
	1	SHORT(A)	MRSSHORT		FASTR1	15	5	1	Dither 1	4	20	876.913	222197
	2		IMAGER	F770W	FASTR1	5	10	1	Dither 1	4	40	654.909	222197
	2	MEDIUM(B)	MRSLONG		FASTR1	15	5	1	Dither 1	4	20	876.913	222197
	2	MEDIUM(B)	MRSSHORT		FASTR1	15	5	1	Dither 1	4	20	876.913	222197
	3		IMAGER	F1000W	FASTR1	5	10	1	Dither 1	4	40	654.909	222197
	3	LONG(C)	MRSLONG		FASTR1	15	5	1	Dither 1	4	20	876.913	222197
	3	LONG(C)	MRSSHORT		FASTR1	15	5	1	Dither 1	4	20	876.913	222197

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 515-Athalia FROM JWST LESS THAN 0.075
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